

Cybersecurity Basics 1 Project, Rubric 1

Create an Incident Response Plan

Develop and submit an incident response plan that outlines at least 1 method for detecting security incidents, 1 strategy for containment, and steps for eradication and recovery and Identifies and explains at least 1 of the types of cyber attacks (Malware, Phishing, Ransomware, and Denial of Service)

Incident Response Plan (IRP)

1. Incident Detection Method

- **Network-Based Intrusion Detection System (NIDS):** Continuous monitoring of network traffic using automated IDS tools (e.g., Snort, Suricata) deployed at key access points. The system inspects incoming and outgoing packet headers and payloads against signatures of known malicious activity and flags anomalous traffic patterns to generate real-time alerts for the security team.

2. Containment Strategy

- **Short-Term Network Isolation:** Immediately sever the affected hosts from the network by disabling physical network interfaces, disconnecting Wi-Fi connections, or placing infected VLANs into an isolated quarantine segment. This stops lateral movement across internal systems while keeping the hardware powered on for forensic memory dumps.

3. Eradication & Recovery Steps

Phase	Action Items
-------	--------------

Eradication

- Identify and terminate all malicious processes running on impacted devices.
- Remove persistence mechanisms (registry keys, scheduled tasks, startup scripts).
- Run complete system scans using updated anti-malware software.
- Apply missing software updates and security patches to eliminate the vulnerability.

Recovery

- Restore affected system configurations and data from clean, verified offline backups.
- Force global password resets and revoke compromised API keys or authentication tokens.
- Perform vulnerability scans to confirm system integrity before reconnecting to the network.
- Re-integrate systems into the production environment under enhanced monitoring.

4. Cyber Attack Analysis: Ransomware

- **Definition:** Ransomware is a form of malware designed to encrypt files or lock access to operating systems.
- **Mechanism:** Once executed on a target system, it encrypts critical files using robust encryption algorithms and displays a ransom demand (often requiring payment in cryptocurrency) in exchange for the decryption key.

- **Impact:** Causes severe operational downtime, potential data loss, and financial or reputational damage. Prevention and response rely on maintaining secure, isolated backups rather than negotiating with threat actors.